

Weather Dashboard - Project Report

Project Overview

The Weather Dashboard is a full-stack Python Django application. It allows users to register, log in, and manage a personalized dashboard of favorite cities. The application fetches real-time weather data and a 5-day forecast using the OpenWeatherMap API.

System Architecture

The architecture follows the MVT (Model-View-Template) pattern: - **Models:** Define the database schema (User, FavoriteLocation, WeatherSearchLog). - **Views:** Handle business logic, utilizing Class-Based Views (CBVs) for CRUD operations. - **Templates:** Render the UI using HTML and Vanilla CSS with a glassmorphism aesthetic. - **Service Layer:** Encapsulates third-party API integration (OpenWeatherMap) to handle data fetching, error handling (404, 429, timeouts), and parsing.

Database Schema

Model	Fields	Relationships / Notes
User	username, password, email	Django built-in auth model
FavoriteLocation	id, user_id, city_name, custom_nickname, key_id	Foreign Key to User. Unique together (user, city)
WeatherSearchLog	id, user_id, query_string, timestamp, status_code	Foreign Key to User. Used for tracking API usage and errors

Error Handling & Validation

- Input Sanitization: User input is cleaned via Django Forms before interacting with the database or API. - Graceful degradation: If the API returns a 404 (City Not Found) or 429 (Rate Limit Exceeded), the service layer raises custom exceptions which are caught by the views and displayed as Django Messages to the user. - Network Timeouts: The requests library is configured with a 5-second timeout to prevent blocking.